

How to Write Problem Statements

These exercises will help kickstart the process for completing the introduction section of your Design Document.

SINGLE SENTENCE PROBLEM STATEMENTS

Write 3 different problem statements using the “How Might We” structure:

1. **How Might We** design an affordable eddy-current probe system **for** engineers measuring the electrical conductivity of metal samples **so that** they can obtain accurate and reliable conductivity measurements.
2. **How Might We** optimize eddy-current probes for different frequency and conductivity ranges **for** researchers and engineers **so that** they can accurately characterize metal materials without direct electrical contact.
3. **How Might We** develop a calibrated conductivity measurement system using eddy-current probes and a VNA **for** students, engineers, and researchers **so that** they can quickly and consistently measure the conductivity of metal samples.

5W1H

Based on your project, answer the following:

WHO has the problem?

Engineers, researchers, and students who need to measure the electrical conductivity of metal samples using a reliable and affordable measurement system.

WHAT is the problem?

There is a need for an affordable conductivity meter that can accurately measure electrical conductivity using eddy-current probes and VNA measurements

WHERE is the problem occurring?

The problem occurs in laboratory and engineering environments where metal materials need to be characterized and their electrical conductivity measured.

WHEN is the problem occurring?

The problem occurs whenever engineers or researchers need to determine the electrical conductivity of metal samples, particularly when different conductivity or frequency ranges require optimized measurement probes

WHY is it important?

Eddy Current (EC) Probes for Conductivity Measurements

The measurement of eddy currents in conductive elements is important because the given output can determine which material or materials the object may be, the purity or quality of the object, and if it fits the required conductivity to be used in a process.

HOW will it be solved?

The problem will be solved by designing and optimizing eddy-current probes, using a vector network analyzer to measure the probes' S-parameters, and calibrating the system with reference samples that have known electrical properties.

PUT IT TOGETHER

Put together your project's problem statement in proper paragraph form.

Engineers, researchers, and students often need to measure the electrical conductivity of metal samples using a reliable and affordable measurement system. The problem here is that conductivity measurement equipment is expensive and may not provide flexibility to test different materials and test ranges. These problems occur in labs and engineering environments whenever the conductivity of a metal sample need to be found. It is also important when testing different materials, evaluating material properties, or comparing different conductivity values. This is an important problem to solve because accurate conductivity measurements help engineers understand material properties, verify material quality, and make informed engineering decisions while keeping equipment costs low as possible. The problem will be addressed by designing eddy current probes that work with an affordable vector network analyzer (VNA). The VNA will measure the S-parameters of the sensor coils, allowing changes in complex impedance caused by the conductivity and permeability of the metal to be observed. The probes will then be optimized for specific frequency and conductivity ranges and calibrated using reference samples with known electrical properties to produce accurate conductivity measurements.